



The Nine Horsemen

CO₂ and CFC Sensors



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INTRODUCTION

CFCs

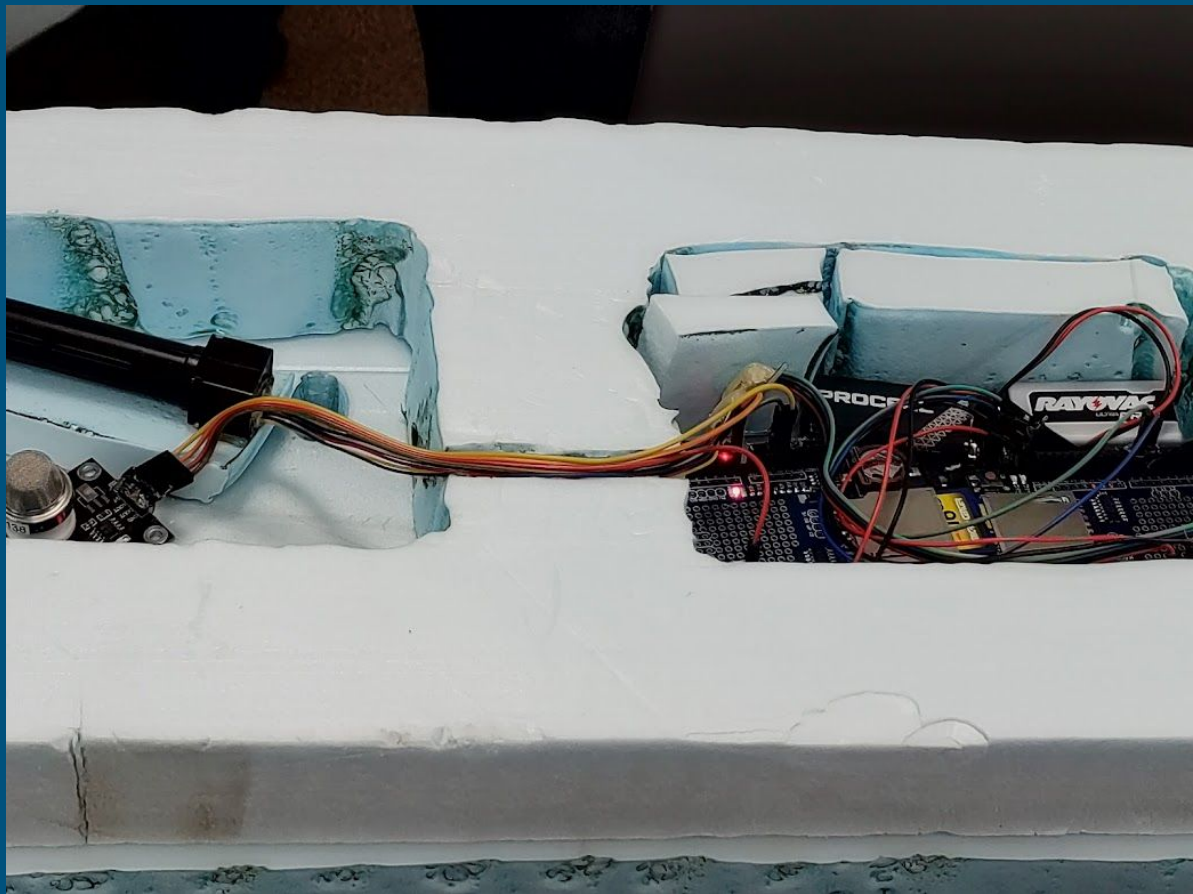
- Location of CFCs in the atmosphere
- We expect most of the CFCs to be in the upper atmosphere because the direction of the wind current is coming from the North West.

CO₂

- Discover where the CO₂ is coming from
- Originally a back-up experiment

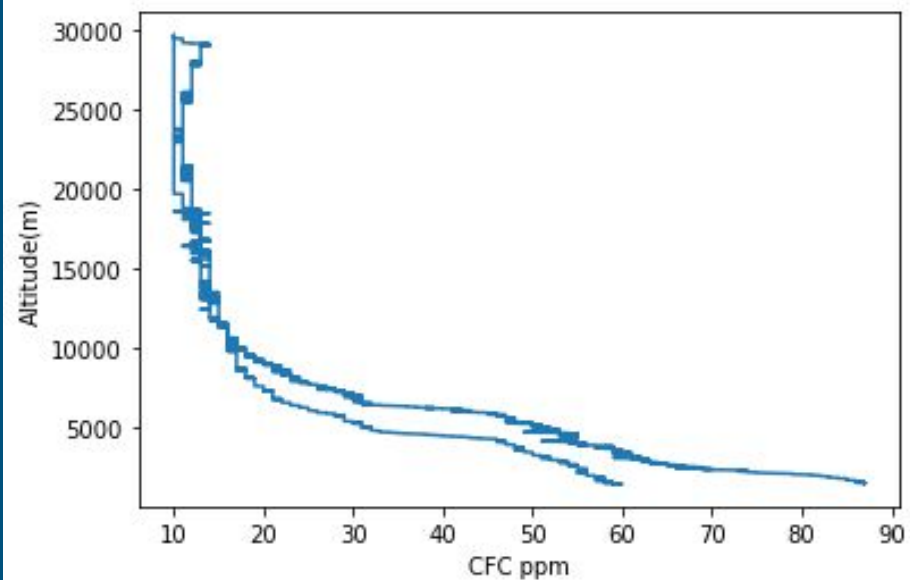
PROCEDURE

- Researching
- Rewiring and Debugging
- Lot and lots of testing
- Prepping for flight
- Day of Flight

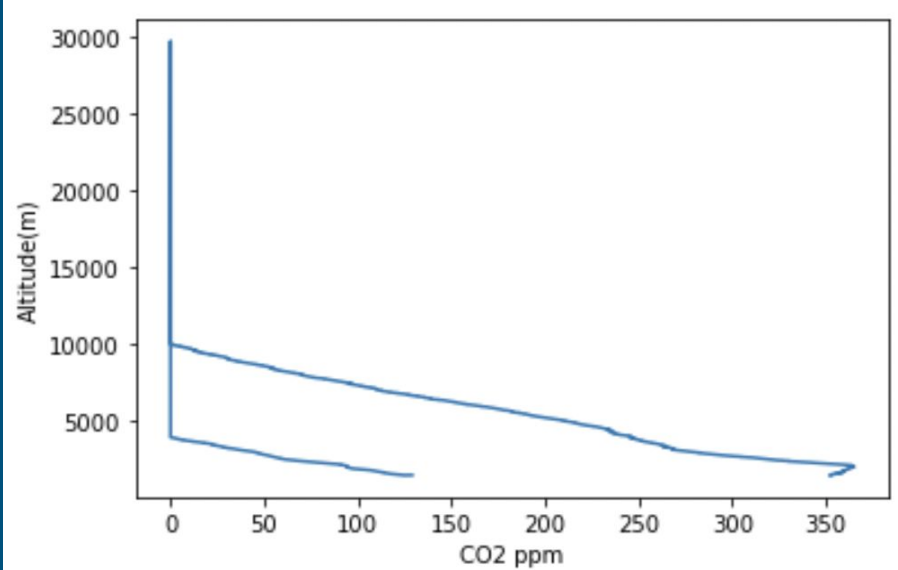


DATA

CFC Data



CO2 Data



ANALYSIS

CFC

- Uncertainties of our measuring device is +/- 50 ppm +/- 5% of the reading

CO2

- Dropping to 0 ppm at about 5000 m was unexpected
- CO2 sensor takes a while to adjust
- +/- 50 ppm +/- 5% of the reading

RESULTS

CFCs

- There were 10-15 ppm CFCs between 15000 and 30000 meters

CO2

- Reads 0 from 10 km to 30 km... very suspicious...
- Too cold for CO2 probe?

CONCLUSION

- Our sensor picked up CFCs in the ozone layer!
- Further analysis and comparison to other CO2 data needed

